SOP: Task 12 – Implement Monitoring Using Azure Monitor

Objective

To implement Azure Monitor for monitoring Azure Virtual Machines, collect performance metrics, analyze resource health, and monitor the infrastructure through Azure Monitor dashboards.

Prerequisites

The following resources were already deployed and available before configuring monitoring:

- Resource Group: **rg-mayank**
- Virtual Network: **vnet-1**
- Virtual Machine: **vm-1**
- Virtual Machine: **vm-2**
- Public IP Address: **vm-1-ip**
- Public IP Address: **vm-2-ip**
- Network Security Groups
- Network Interfaces
- Managed Disks
- Azure Subscription

Step 1: Verify Existing Resources

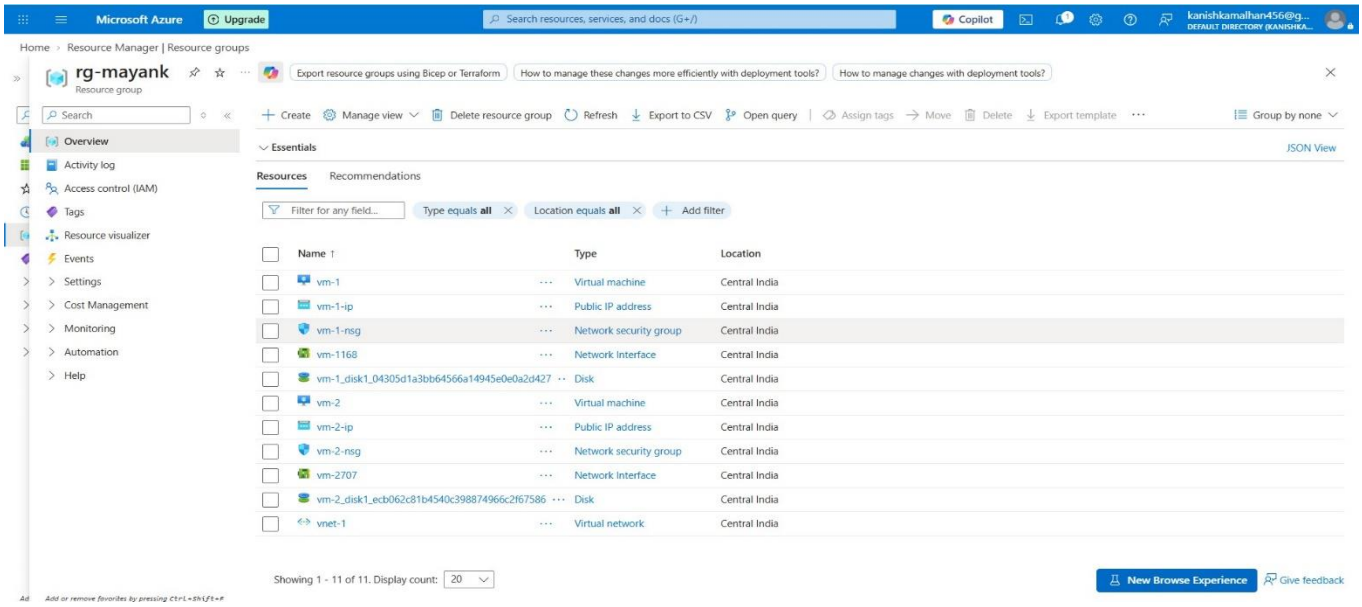
Navigate to:

Azure Portal
→ Resource Group
→ rg-mayank

Review the deployed resources and verify that all required infrastructure components are available.

Observation

The Resource Group contains the virtual machines, public IP addresses, network interfaces, network security groups, managed disks, and virtual network required for monitoring.



Step 2: Open Azure Monitor

Navigate to:

→ Monitor

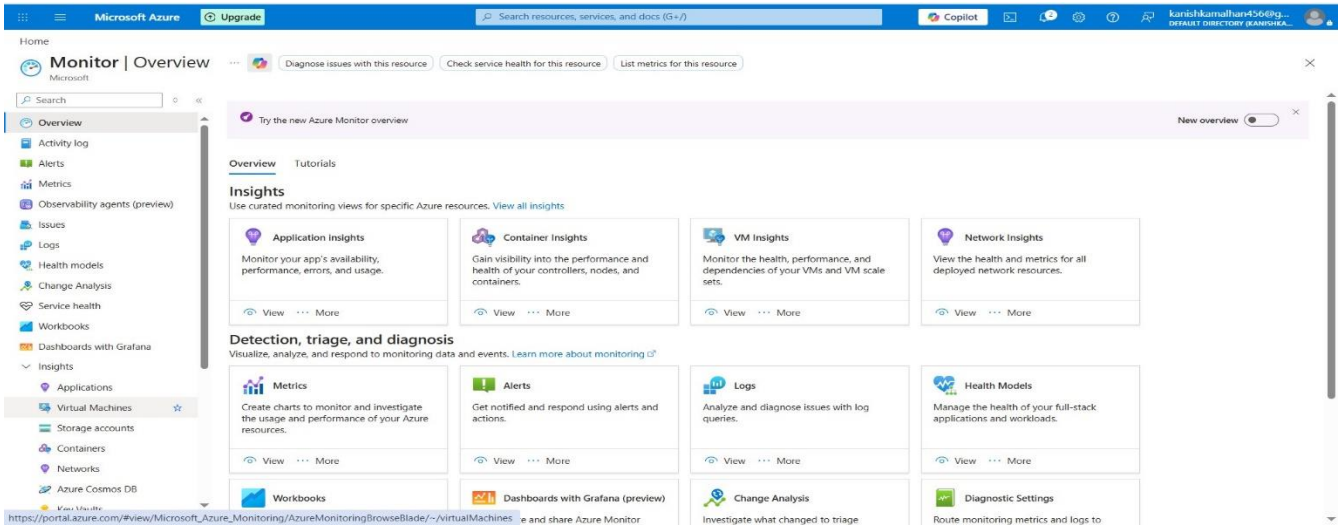
Azure Monitor provides centralized monitoring and observability services for Azure resources.

Available monitoring services include:

- VM Insights
- Metrics
- Logs
- Alerts
- Network Insights
- Application Insights

Observation

Azure Monitor dashboard was successfully opened.



Step 3: Access Virtual Machine Monitoring

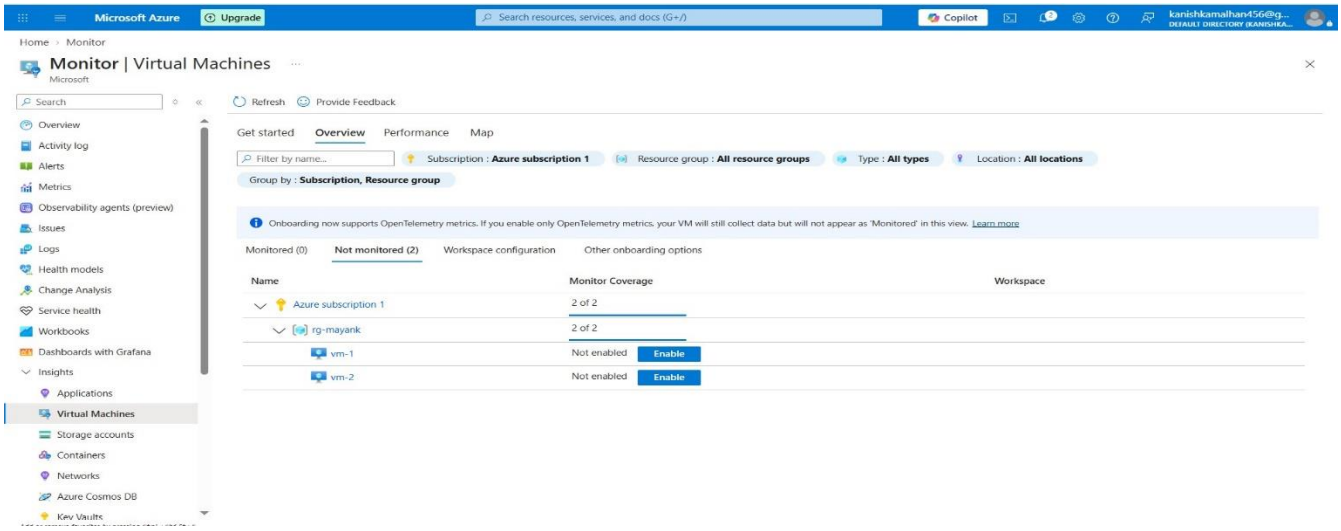
Navigate to:

Monitor
→ Insights
→ Virtual Machines

Azure Monitor automatically discovers available virtual machines.

Observation

The virtual machines **vm-1** and **vm-2** were detected, however monitoring was not enabled.



Step 4: Configure Monitoring for Virtual Machines

Click **Enable** for the virtual machine to configure Azure Monitor.

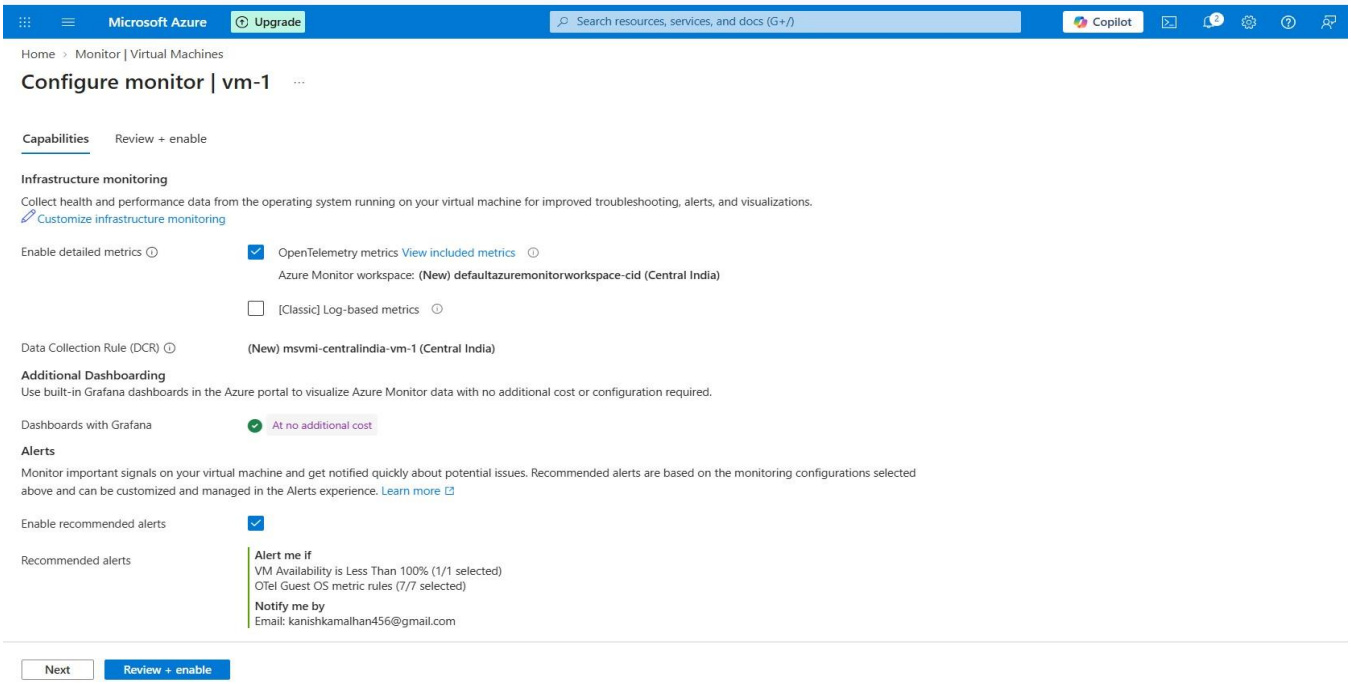
Navigate to:

Monitor
→ Virtual Machines
→ vm-1
→ Enable

A Data Collection Rule (DCR) is automatically created during configuration.

Observation

Azure Monitor onboarding configuration was successfully prepared for metric and telemetry collection.



Step 5: Enable Azure Monitor

Select:

Review + Enable

Azure automatically performs the following tasks:

- Deploys Azure Monitor Agent
- Creates Data Collection Rules (DCR)
- Associates Azure Monitor Workspace
- Configures telemetry and metric collection
- Creates recommended monitoring configurations

Observation

Monitoring onboarding was successfully initiated for the virtual machines.

Step 6: Verify Monitoring Status

After onboarding is completed, navigate to:

Monitor
→ Virtual Machines

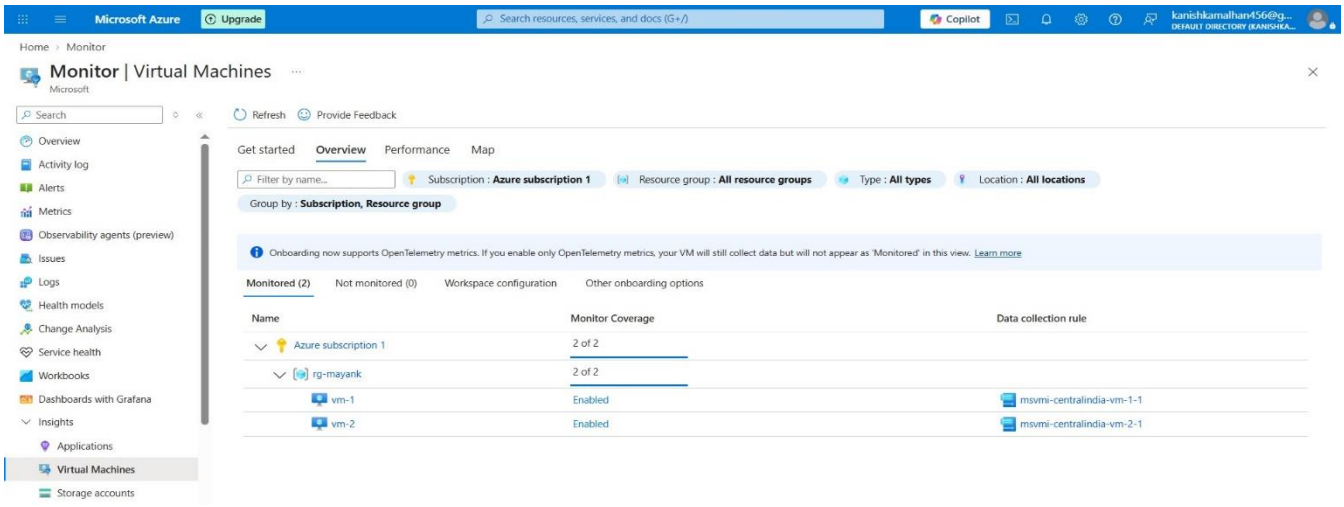
Verify the monitoring status of both virtual machines.

Observation

Monitoring was successfully enabled for both VMs.

Plain Text
vm-1 → Enabled
vm-2 → Enabled

Azure Monitor also created and associated Data Collection Rules for both virtual machines.



Step 7: Verify Performance Metrics Collection

Navigate to:

Monitor
→ Virtual Machines
→ Performance

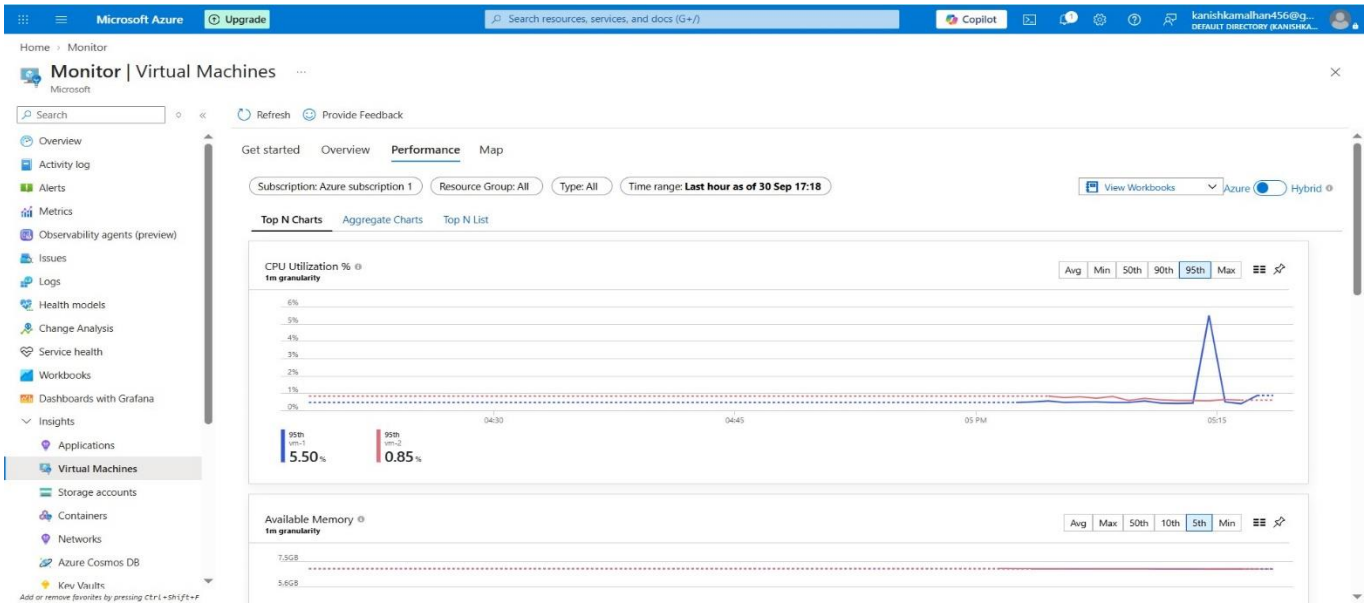
Review the performance charts generated by Azure Monitor.

Metrics available include:

CPU Utilization
Available Memory
Performance Statistics
Resource Utilization Trends

Observation

Azure Monitor successfully collected and displayed performance metrics from both virtual machines. CPU utilization and memory metrics were visible through the monitoring dashboard.



Step 8: Verify Monitoring Resources Using Resource Visualizer

Navigate to:

Resource Group

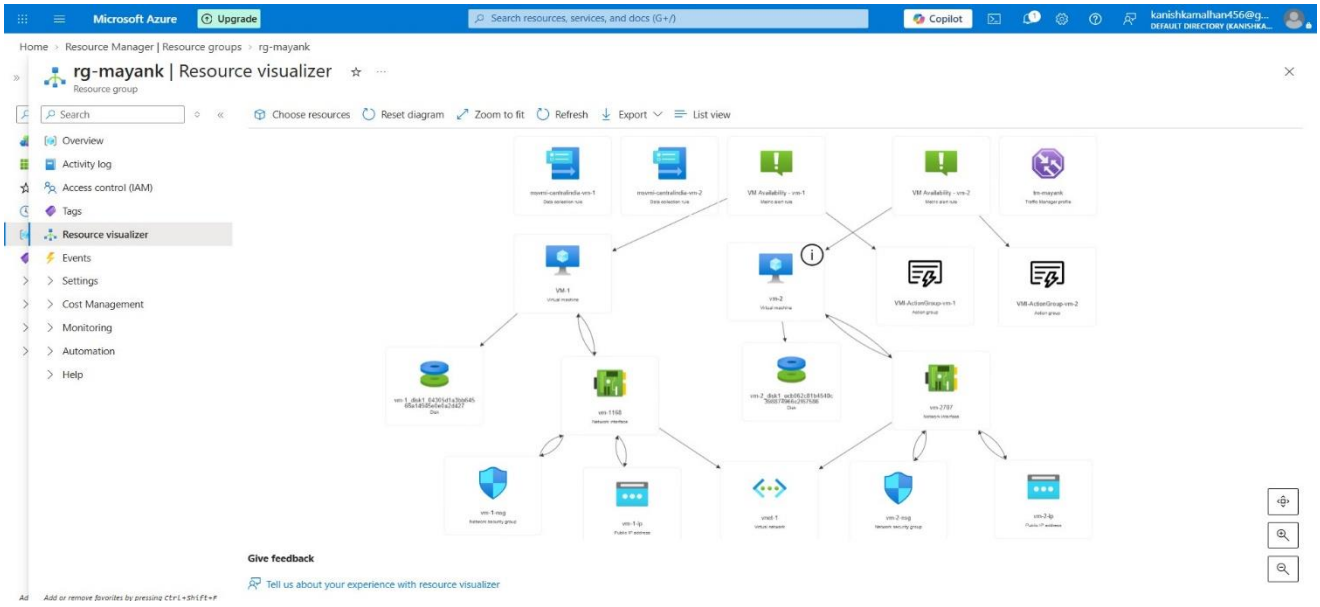
→ rg-mayank

→ Resource Visualizer

Review the resource dependency diagram generated by Azure.

Observation

The Resource Visualizer confirmed that Azure Monitor was successfully integrated with the virtual machines. Data Collection Rules, Alert Rules, and Action Groups were automatically associated with the monitored virtual machines, validating successful monitoring configuration.



Final Observation

Azure Monitor successfully detected and onboarded both virtual machines (**vm-1** and **vm-2**) for monitoring. Azure Monitor Agents and Data Collection Rules were deployed automatically, enabling collection of telemetry and performance data. The monitoring dashboard displayed resource performance statistics including CPU utilization and memory metrics. The Resource Visualizer further confirmed the successful association of monitoring components such as Data Collection Rules, VM Availability Alert Rules, and Action Groups with the monitored infrastructure.

Result

Successfully implemented **Azure Monitor** for the Azure infrastructure hosted in **rg-mayank**. Monitoring was enabled for both virtual machines (**vm-1** and **vm-2**), telemetry collection was configured using Azure Monitor Agent and Data Collection Rules, performance metrics were visualized through Azure Monitor dashboards, and monitoring resources were validated using Resource Visualizer. The environment is now capable of collecting, analyzing, and monitoring virtual machine health and performance in real time.